Sinkhorn algorithm

same as entropic OT!

Sinkhorn Divergence

$$S_\varepsilon(\alpha,\beta) = \mathrm{OT}_\varepsilon(\alpha,\beta) - \frac{1}{2}\mathrm{OT}_\varepsilon(\alpha,\alpha) - \frac{1}{2}\mathrm{OT}_\varepsilon(\beta,\beta)$$

Computational Complexity of SD

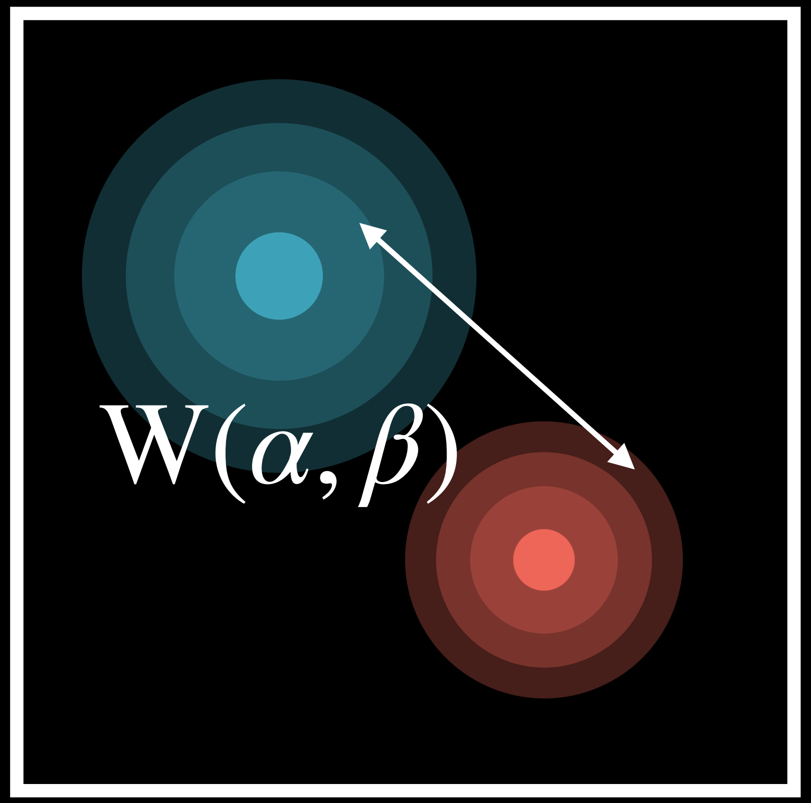
Sinkhorn Divergence

$$S_\varepsilon(\alpha,\beta) = \mathrm{OT}_\varepsilon(\alpha,\beta) - \frac{1}{2}\mathrm{OT}_\varepsilon(\alpha,\alpha) - \frac{1}{2}\mathrm{OT}_\varepsilon(\beta,\beta)$$

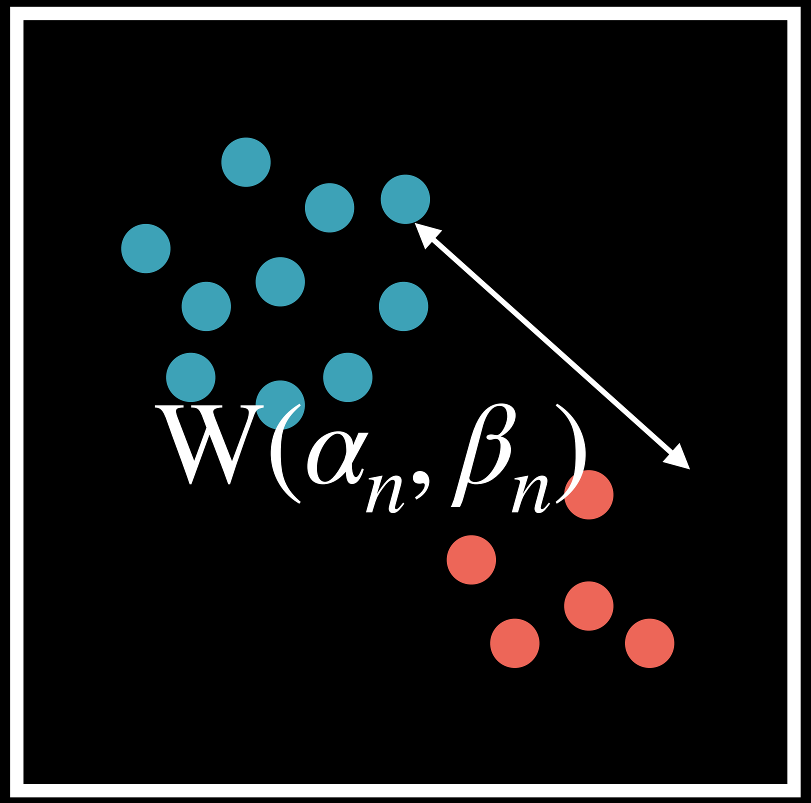
- Compute each term with **Sinkhorn algorithm**
- Can compute the three terms in parallel
- Self-distances empirically faster to compute, easy to initialize well (why?)
- Complexity: $\mathcal{O}(n^2/\varepsilon^2)$ (**same as entropic OT!**)

Side Note: Sample Complexity

If we estimate



with



, how large should n be?